Xiaojia Wang

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Professional Appointments

Postdoctoral Assistant Professor **2022 – 2025**
Department of Mathematics, University of Michigan, Ann Arbor, MI *Mentor: Prof. Silas Alben*

Education

Purdue University, West Lafayette **2018 – 2022**
Ph.D. in Mechanical Engineering *Advisor: Prof. Ivan C. Christov*
Dissertation: Modeling and Stability of Flows in Compliant Microchannels

Shanghai Jiao Tong University, China **2015 –2018**
M.S. in Naval Architecture and Ocean Engineering *Advisor: Prof. Renchuan Zhu*

Tianjin University, China **2011 – 2015**
B.S. in Ocean and Naval Engineering
Minor in English Literature

Research Interests

Fluid mechanics | Microfluidics | Biological flows | Fluid–structure interactions | Hydrodynamic stability | Vortex dynamics | Scientific computing | Mathematical modeling

Awards and Honors

- **Conference Travel Grant** from the College of Engineering, Purdue University, 2021.
- **Rising Stars in Mechanical Engineering,** 2021. Selected as one of the (around) 30 participants in the United States attending the Academic Career Workshop for Women in Mechanical Engineering at Massachusetts Institute of Technology, Cambridge, MA.
- **Ross Fellowship** from The Graduate School at Purdue University, 2018
- **Second Prize** in China Post-Graduate Mathematical Contest in Modeling, 2017
- **First-class Academic Scholarship** from Shanghai Jiao Tong Univeristy, 2015
- **Puxin Environmental Protection Scholarship** from Puxin Co. Ltd, 2015
- **CCS scholarship** at Tianjin University, 2012, 2013
- **Three A's Student** at Tianjin University, 2012 – 2015

Funded Projects

1. **Optimal Flows for Thermal Transport** **2022–**

   Sponsor: U.S. National Science Foundation–Division of Mathematical Sciences

   Principle Investigator: Prof. Silas Alben

2. **Nonlinear Dynamics of Confined Interfaces: Beyond Linear Analysis and Towards Control** **2020–2024**

   Sponsor: U.S.National Science Foundation–Division of Civil, Mechanical, and Manufacturing Innovation

   Principle Investigator: Prof. Ivan C. Christov

3. **Microscale Fluid–Structure Interactions: Towards a Predictive Theory of Their Dynamic Response** **2017–2022**

   Sponsor: U.S.National Science Foundation–Division of Chemical, Bioengineering, Environmental, and Transport Systems

   Principle Investigator: Prof. Ivan C. Christov

Publications

Google Scholar

In Refereed Journals:

1. S Alben, **X Wang**, N Vuong, Enhancing wall-to-wall heat transport with unsteady flow perturbations, *Journal of Fluid Mechanics*, 1024 A51, 2025. doi: 10.1017/jfm.2025.10931

2. **X Wang**, S Alben, How vortices enhance heat transfer from an oscillating plate, *Journal of Fluid Mechanics*, 1013 A47, 2025. doi: 10.1017/jfm.2025.10201

3. S Pande, **X Wang**, IC Christov, Oscillatory flows in compliant conduits at arbitrary Womersley number, *Physical Review Fluids*, 8: 124102, 2023. doi: 10.1103/PhysRevFluids.8.124102

4. **X Wang**, S Pande, IC Christov, Flow rate–pressure drop relations for new configurations of slender compliant tubes arising in microfluidics experiments, *Mechanics Research Communications*, 126: 104016, 2022. doi: 10.1016/j.mechrescom.2022.104016

5. **X Wang**, IC Christov, Reduced modeling and global instability of finite-Reynolds-number flow in compliant rectangular channels, *Journal of Fluid Mechanics*, 950 A22, 2022. doi: 10.1017/jfm.2022.802

6. **X Wang**, IC Christov, Reduced models of unidirectional flows in compliant rectangular ducts at finite Reynolds number, *Physics of Fluids*, 33(10):102004, 2021. doi: 10.1063/5.0062252

   ☞ Featured Article. Contribution to the special issue Tribute to Frank M. White on his 88th Anniversary.

7. TC Inamdar, **X Wang**, IC Christov, Unsteady fluid-structure interactions in a soft-walled microchannel: A one-dimensional lubrication model for finite Reynolds number, *Physical Review Fluids*, 5(06): 064101, 2020. doi:10.1103/PhysRevFluids.5.064101

   ☞ Conducted the scaling analysis and developed the numerical method. The contributed part was included as one chapter of X Wang's PhD thesis.

8. Q Xiao, R Zhu, S Huang, **X Wang**, Responses of ship motion to nonlinear focusing waves based on high-order spectral method, *Shipbuilding of China*, 61(01): 50-59, 2020. (In Chinese) doi: CNKI:SUN:ZGZC.0.2020-01-005

9. **X Wang**, IC Christov, Theory of the flow-induced deformation of shallow compliant microchannels with thick walls, *Proceedings of the Royal Society A*, 475(2231): 20190513, 2019. doi: 10.1098/rspa.2019.0513

10. H Cai, R Zhu, **X Wang**, J Fan, Dynamic response analysis of floating offshore wind turbine in combined wind and wave, *Journal of Harbin Engineering University*, 40(01): 118-125, 2019. (In Chinese) doi: 10.11990/jheu.201709043

11. **X Wang**, R Zhu, L Hong, Kramers–Kronig relations and frequency to time-domain transformation method for time domain calculation of floating body with forward speed, *Shipbuilding of China*, 59(2): 9-15, 2018. (In Chinese) doi: CNKI:SUN:ZGZC.0.2018-02-002

In Conference Proceedings:

- **X Wang**, IC Christov, Soft hydraulics in channels with thick walls: The finite-Reynolds-number base state and its stability, *AIP Conference Proceedings*, 2302: 020002, 2020. doi: 10.1063/5.0033517
- **X Wang**, R Zhu, L Hong, Time domain calculation of floating body with forward speed in waves and derivation and validation for unified expressions of Kramers–Kronig relations, *Proceedings of the 14th National Congress on Hydrodynamics and the 28th National Conference on Hydrodynamics*, 2017. (In Chinese)

Teaching and Mentorship

University of Michigan — **Instructor**

- Math 115 Calculus I (two sections) — Fall 2024
- Math 115 Calculus I (**Faculty lead**, one section) — Fall 2023
- Math 115 Calculus I (two sections) — Winter 2023
- Math 115 Calculus I (two sections) — Fall 2022

Comments from students:

"The instructor is very thorough when she teaches and wants to make sure we understand everything."

"The continuing of group work should be encouraged. I very much appreciate that Xiaojia is supportive of this group culture, not only in team homework, but also within class and on problems. She cares a lot about the class and treats everyone with great respect."

"While concepts are at times confusing, the instructor does her best to clear up any misunderstandings and is always happy to answer any questions that we may have. The classroom environment is very good; everyone gets along really well and collaboration is both encouraged and generally helpful for learning. "

"Dr. Wang is very passionate about the subject and at times can make learning the content fun. Sometimes it can be hard to understand the topic but in the end, after several examples I can piece together and have enough of an idea to know the rest."

University of Michigan **Research Mentor**

summer 2024

- Co-advised a student (with Prof. Silas Alben), Nicole Vuong, under the Research Experience for Undergraduates (REU) program performing an independent study on **Enhancing wall-to-wall heat transfer by unsteady flow perturbations**.
- Held regular meeting with the student. Helped with the derivations and numerical schemes.

Purdue University **Research Mentor**

Spring 2021 – Summer 2022

- Advised a Purdue Mechanical Engineering master student, Shrihari Pande, performing an independent study on **modeling and numerical simulations of oscillatory flows in a compliant tube.**
- Advised a Purdue Mechanical Engineering undergraduate student, Shrihari Pande, performing an independent study on **modeling and stability in compliant rectangular microchannels**.
- Prepared educational learning materials (references, codes, etc.). Supervised progress on research.

Shanghai Jiao Tong University **Teaching Assistant**

- For Lab in School of SJTU-Paris Tech Elite Institute of Technology. Sept. 2016 – Jun. 2017
- For Course: *Potential theory of ship motion in waves.* Mar. 2016 – Jun. 2016

Leadership and Service

Peer Review

- Journal of Fluid Mechanics
- Physics of fluids

Student Volunteer **Jul. 2016**

- In 2016 International Summer School on Naval Architecture, Ocean Engineering and Mechanics at Shanghai Jiao Tong University.
- Assisted in organizing seminars.
- Provided guidance for international speakers.

Member of Summer Practice Team, Tianjin University **Jul. 2013**

- Investigated the state of education in the northwestern part of China.
- Taught English for students from the first to the third grade in Lan Lian Primary School, Gansu Province, China.

Selected Conference Presentations

- **X Wang**, SD Pande, IC Christov, Nonlinear flow rate–pressure drop relations in slender compliant microtubes, The 75th Annual Meeting of the APS DFD, Indianapolis, USA, 2022.
- **X Wang**, Reduced modeling and global instability of low-Reynolds-number flows through compliant microchannels, Engineering and Applied Science Forum Webinar, virtual, 2022.
- **X Wang**, IC Christov, Reduced models for analyzing flow instability in compliant rectangular microchannels, The 74th Annual Meeting of the APS DFD, Phoenix, USA, 2021.
- **X Wang**, IC Christov, Modeling and stability in compliant microchannels, SPARC Workshop on Fluidics involving deformable interfaces, virtual, 2021.
- **X Wang**, IC Christov, Theory of the bulging effect of soft microchannels with thick walls, The 72nd Annual Meeting of the APS DFD, Seattle, USA, 2019.
- **X Wang**, R Zhu, L Hong, Time domain calculation of floating body with forward speed in waves and derivation and validation for unified expressions of Kramers–Kronig relations, The 28th National Conference on Hydrodynamics, Changchun, China, 2017.

Professional Development

- The 2021 SES (Virtual) Month: Mechanics Matters, virtual, Oct. 15, 2021. Participated in the online poster session, presenting **Dancing microchannels: Reduced models for unsteady fluid–structure interactions at the microscale**.
- XSEDE HPC Workshop Summer Boot Camp, virtual, Jun. 8–11, 2021.
- Six-week short course on *Intermediate Pronunciation and Prosody*, Purdue Language and Cultural Exchange, Aug. 31– Oct. 07, 2020.
- Mathematical Fluids, Materials and Biology Workshop, Ann Arbor, USA, Jun. 13–15, 2019. Participated in poster session, presenting **Towards a theory of fluid–structure interaction due to internal flow in deformable microchannels**.
- The 45th Midwest Universities Fluid Mechanics Retreat (MUFMECH 2019), Rochester, Indiana, USA, Apr. 11–13, 2019.
- Software Carpentry Workshop, Purdue University, Oct. 8–10, 2018.
- Summer internship at Hudong-Zhonghua Shanghai Shipyard, Shanghai, China, Jul. 2014.

Expertise and Skills

- Programming: Python, Matlab, FORTRAN, LaTeX, Mathematica
- High Performance Computing: MPI, Slurm, Bash
- Finite element analysis: svFSI of SimVascular, FEniCS, FreeFEM

Languages

- **English**: Fluent; Test for English Majors Grade 8 (TEM8); Test for English Majors Grade 4 (TEM4)
- **Mandarin**: Native